MILITARY SPECIFICATION

# HEPARIN SODIUM INJECTION, U.S.P.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers Heparin Sodium Injection, U.S.P.

1.2 Classification. The injection covered by this specification shall be of the following sizes and strengths (See 6.2):

Strength 1 -1,000 Units per ml
Strength 2 -5,000 Units per ml
Strength 3 -10,000 Units per ml
Strength 4 -20,000 Units per ml

Size 1 - 1 ml, 10s
Size 2 - 2 ml, 12s
Size 3 - 5 ml
Size 4 - 10 ml

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

FEDERAL

FED-STD-142 - Parenteral Preparations.



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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Heparin Sodium Injection, U.S.P., and shall be in accordance with all applicable requirements of FED-STD-142, and as specified herein.



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S2. Classification. Shall be type I, class 1.

S5.2.1 The following is added to this paragraph:

Strength 1 shall contain 1,000 Units of Heparin per ml, within the applicable assay limits for the injection.

Strength 2 shall contain 5,000 Units of Heparin per ml, within the applicable assay limits for the injection.

Strength 3 shall contain 10,000 Units of Heparin per ml, within the applicable assay limits for the injection.

Strength 4 shall contain 20,000 Units of Heparin per ml, within the applicable assay limits for the injection.

3.2 Ingredients.

3.2.1 Heparin Sodium. The Heparin Sodium used in the manufacture of the injection shall comply with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2.2 Other ingredients. All other ingredients used in the manufacture of the injection shall comply with S5.1 of FED-STD-142.

3.3 Expiration dating period. The expiration dating period shall be 60 months for size 1 and not less than 36 months for sizes 2, 3 and 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-140.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One box containing 10 cartridge-needle units of strength 3, size 1, or one box containing 10 cartridge-needle units of strength 2, size 1, or one bottle containing 10 ml of strength 1, size 4, or one package containing 12 vials of strength 4, size 2, or one bottle containing 5 ml of strength 3, size 3, as specified constitutes one unit of issue.

5.2.2 Level A.



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5.2.2.1 Immediate container.

5.2.2.1.1 Strength 3, size 1. The immediate container for strength 3, size 1, shall be a commercially available disposable cartridge-needle unit, with 25 gage X 5/8 inch needle, considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The cartridge-needle unit shall be compatible with 6515-00-926-9043 Syringe, Cartridge.

5.2.2.1.2 Strength 2, size 1. The immediate container for strength 2, size 1, shall be a commercially available disposable cartridge-needle unit, with 22 gage x 1-1/4 inch needle, considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The cartridge-needle unit shall be compatible with 6515-00-926-9043 Syringe, Cartridge.

5.2.2.1.3 Strength 1, size 4, and strength 3, size 3. The immediate container for strength 1, size 4, and strength 3, size 3, shall be a commercially available immediate container considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.1.4 Strength 4, size 2. The immediate container for strength 4, size 2, shall be a commercially available vial considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. Each unit of strength 3, size 1, or strength 2, size 1, or strength 4, size 2, shall be packaged as specified in 5.2.5 of PPP-C-186. At the option of the contractor, each unit of strength 1, size 4 or strength 3, size 3, may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twenty (20) units of issue of strength 3, size 1, or strength 2, size 1, shall be packaged as specified in 5.2.7.1 of PPP-C-186. Twelve (12) units of issue of strength 1, size 4, or ten (10) units of issue of strength 3, size 3, shall be packaged as specified in 5.2.7.1 of PPP-C-186. Twelve (12) units of issue of strength 4, size 2, shall be packaged as specified in 5.2.8 of PPP-C-186.



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5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred twenty (120) units of issue of strength 3, size 1, or strength 2, size 1, or four hundred thirty-two (432), units of issue of strength 1, size 4, or seventy-two (72) units of issue of strength 4, size 2, or four hundred (400) units of issue of strength 3, size 3, shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred twenty (120) units of issue of strength 3, size 1, or strength 2, size 1, or four hundred thirty-two (432) units of issue of strength 1, size 4, or four hundred (400) units of issue of strength 3, size 3, shall be packed as specified in 5.4.3.1.1 of PPP-C-186. Seventy-two (72) units of issue of strength 4, size 2, shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.



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NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied for sizes 3 and 4, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legends:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"
AND
"DO NOT PERMIT TO FREEZE"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I Shelf life marking as specified MIL-STD-129 shall also be shown. Marking shall include the legends:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"
AND
"DO NOT PERMIT TO FREEZE"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):



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| National Stock Number | Item Identification | Specification Designation |
|---|---|---|
| 6505-00-133-5496 | HEPARIN SODIUM INJECTION, USP, 10,000 Units per ml, Cartridge-Needle Unit, 1 ml, 10s | Strength 3, size 1 |
| 6505-00-133-5497 | HEPARIN SODIUM INJECTION, USP, 5,000 Units per ml, Cartridge-Needle Unit, 1 ml, 10s | Strength 2, size 1 |
| 6505-00-153-9740 | HEPARIN SODIUM INJECTION, USP, 1,000 Units per ml, 10 ml | Strength 1, size 4 |
| 6505-00-579-8432 | HEPARIN SODIUM INJECTION, USP, 20,000 Units per ml, 2 ml, 12s | Strength 4, size 2 |
| 6505-00-584-2914 | HEPARIN SODIUM INJECTION, USP, 10,000 Units per ml, 5 ml | Strength 3, size 3 |

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1594



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-H-37456 HEPARIN SODIUM INJECTION, U.S.P.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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